# José Manuel Requena Plens

## Firmware & Software Engineer Firmware • Software • QA

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**Embedded firmware and software engineer** with experience building industrial firmware for solar inverters (C, STM32, FreeRTOS) and a solid R&D background in acoustics (physics). Focused on low-level optimization, industrial communications (Modbus) and hardware test automation. I maintain *open source* projects used by developers worldwide. Targeting **firmware, software and QA** roles, bringing engineering rigor and a focus on quality.

On-site, remote or hybrid

## // EXPERIENCE

### R&D Software Developer
Apr 2023 - May 2026

Power Electronics S.L. · Solar + Storage · Lliria, Spain

*Firmware development for solar inverters and control systems in photovoltaic plants, working on STM32 microcontrollers with bare-metal and RTOS environments.*

- Reduced RTOS task stack usage by 30% to 50%, optimizing memory on STM32 microcontrollers.
- Optimized Modbus TCP request handling, cutting CPU usage under load (40,000 req/s) from 80% to 17% with no dropped responses.
- Redesigned embedded file-system access, cutting access times by over 50%.
- Implemented a scheduled purge (daily/monthly) that safely clears thousands of files, preventing storage exhaustion under continuous operation.
- Built and maintained industrial communications in C with hardware abstraction layers (HAL): Modbus RTU/TCP, UART, I2C and SPI.
- Automated the test suite: ~50 to 80 unit tests per module (Ceedling, thousands in total) reaching 100% line and branch coverage via standard-library mocks/fakes, plus hundreds of on-hardware integration tests (pytest) for TCP/FTP performance and FTP security, all in CI.
- Reviewed merge requests across a 12-person team, catching bugs and proposing improvements before integration.

### Predoctoral researcher
Jan 2023 - Apr 2023

Polytechnic University of Valencia + Spanish National Research Council + Hospital La Fe · Medical Imaging and Therapy Systems · Valencia, Spain

*Researcher in the R&D line: **Acoustic Metamaterials for Ultrasound Histotripsy treatments**.*

- Ultra-close focus acoustic lens design.
- Prototype manufacturing.
- Experimentation with ex-vivo tissues.

### Predoctoral researcher
Jul 2021 - Jul 2022

Polytechnic University of Valencia + Spanish National Research Council · Medical Imaging and Therapy Systems · Valencia, Spain

*The main objective is to conduct research and development of acoustic metamaterials for use in architectural applications and medical imaging.*

- Developed acoustic metadiffusers reaching a diffusion coefficient of 0.8 with a thickness 10× smaller than traditional diffusers.
- Simulation of the acoustic diffusers and validation of results (MATLAB & COMSOL).
- Designed and 3D-printed (resin) slitted-tube waveguides with numerically-computed slits (2 to 3 cm tubes, ~0.5 mm slits) that increased ultrasound directivity and range ~15× in air, validated experimentally.
- Validated experimental results against those obtained in simulation.
- Advisor of the Master thesis of a student of the Master's Degree in Acoustic Engineering of the UPV.

## TECHNICAL SKILLS

**Embedded Firmware & Electronics**
C
STM32 & ARM
ESP32 / ESP-IDF
PlatformIO
Texas Instruments
FreeRTOS / RTOS
Comm. Protocols
HW Debugging
Keil uVision
Instrumentation
Secure Boot & Embedded Security
Doxygen

**Software Engineering & Tools**
Python
C++
Go
TypeScript
Bash
Clang Tooling
Git
GitHub
GitLab
Docker
MATLAB
CMake
JetBrains Toolbox

**Quality, CI/CD & DevSecOps**
GitHub Actions
GitLab CI
Jenkins
SonarQube / SonarCloud
GoReleaser
Static Analysis & SAST
Testing
pre-commit hooks

**Networking**
TCP/IP & Networking
MikroTik RouterOS
WireGuard / VPN
IPv6 & Dual-stack
QUIC / HTTP3
DNS / Dynamic DNS

**Security & Cryptography**
Applied Cryptography
TLS / mTLS / PKI
Nginx Hardening
CrowdSec
Network Hardening

**AI & Agentic Engineering**
Model Context Protocol (MCP)
AI Coding Agents
LLM Integration

**Simulation & Acoustics**
COMSOL Multiphysics
LabView

### Researcher in European project
Feb 2020 - Jul 2021

Higher Polytechnic School of Gandia + European Space Agency · Complex Media Acoustics Group of IGIC · Gandia, Spain

*Implement and experimentally validate a sound mitigation method, applicable to a real space vehicle launch configuration (VEGA), that results in a significant decrease in sound pressure levels generated in the launch area during spacecraft liftoff.*

- Designed the optimum acoustic-metamaterial geometry, achieving near-total absorption ($\alpha \approx 1$) across 200 to 500 Hz with a subwavelength design (wavelength/thickness ratio ≈ 16.5 vs ~2 conventional), validated on real lab prototypes.
- Simulation of a real environment to validate design performance (MATLAB, COMSOL & Python).
- Designed and 3D-printed the first prototype; for the large-scale version designed the injection-molding model and sourced the manufacturing of 600 modules with a supplier.
- Development of software for acoustic measurements according to ISO 10534-2 and ASTM E2611 standards. A|Lab
- Design and development of a 3-axis robotic system to perform experimental measurements. Photos and some information here: NEWS
- Results validated and accepted by ESA.

### Research internship
Feb 2018 - Jul 2018

University of Alicante · Dep. of Physics, Systems Engineering and Signal Theory · Alicante, Spain

*Internship in different research projects in the Applied Acoustics Group, belonging to the University Institute of Physics Applied to Sciences and Technologies of the Higher Polytechnic School of Alicante.*

- Simulation of acoustic models.
- Acoustic measurements by means of a LabView-controlled robotic system.
- Investigation of the acoustic radiation efficiency (Vibration-borne noise) of metallic plates for an initial study of ship noise emissions in water. Carried out in collaboration with SAES, here you can read a news item on the subject: NEWS.

### Founder and Vocal Member
Sep 2016 - Jul 2018

AmicroTech · University of Alicante · Alicante, Spain

*Association based at the Higher Polytechnic School of the University of Alicante. Promotes project-oriented microcontroller programming knowledge.*

- Taught 30+ students in reinforcement and extension classes for the subject "Digital Electronic Systems" of the Degree in Sound and Image Engineering of the University of Alicante.
- Laboratory organization and planning (located at the Colegio Mayor).
- Communication and Public Relations.

### Sound technician
Jun 2011 - Feb 2018

Acusticox · Cox, Spain

- Equipment installation and adjustment for events.
- Equipment installation and adjustment for permanent installation.
- FOH Technician.
- Customer acquisition.

### Laboratory technician
Sep 2005 - Nov 2009

Profdent · Alicante, Spain

- Reception of work and casting of plaster models from impressions (alginate/silicone), with and without implants.
- Mounting articulators and casting metal crowns in a centrifuge.
- Ceramic application on crowns and bridges.

## // OPEN SOURCE PROJECTS

**gitlab-mcp-server** — Go · Model Context Protocol server

47 ★ • 59 releases • ~170k downloads • On npm, PyPI & NuGet; listed on Glama, mcp.so, Cursor & mcpservers.org

Acoustic Instrumentation ●●●●○

### General & Platforms
Linux / Unix ●●●●○
MacOS ●●●●○
Windows ●●●●○
LaTeX ●●●●○
HTML / CSS ●●●○○

## LANGUAGES

**Spanish** — Native
**English** — Professional

## CERTIFICATES

### Occupational Health and Safety
Generic (INVASSAT, 50 h)
Nanomaterials (INVASSAT, 50 h)
Chemical sector (INVASSAT, 50 h)
Emergencies (INVASSAT, 70 h)
Nutrition (INVASSAT, 50 h)
Educational (INVASSAT, 50 h)
Services (INVASSAT, 50 h)
Researcher (UPV, 15 h)

### Transversal competences
Gender perspective (EVES, 20 h)
Teamwork (Labora, 25 h)
Design Thinking (Labora, 25 h)
Critical Thinking (Labora, 25 h)
Adaptability, flexibility and agility (Labora, 25 h)
Workplace autonomy and innovation, initiative and creativity (Labora, 25 h)
Improving professional efficiency (Labora, 25 h)
Entrepreneur gender perspective (UPV, 20 h)

### Labor/Industrial
Op. Industrial trucks (Gescoform, 15 h)
Food Hygiene Course (Asonoman, 30 h)
Self-protection plans (INVASSAT, 15 h)
Static electricity (INVASSAT, 15 h)
Data privacy (CSIRT-CV, 10 h)

### Information technology
Using Python for Research (HarvardX, 50 h)
Analyzing Data With Python (IBM, 20 h)
Visualizing Data with Python (IBM, 20 h)

## PUBLICATIONS

### Journal articles
Loudspeakers for Vented Enclosures: a Backwards Approach for Speaker Selection (LoVE BASS) · *Voice Coil*, 2019

### Conference papers
Perfect broadband sound absorber metamaterial for noise reduction in a rocket launch · *European Congress and Exposition on Noise Control Engineering (EuroNoise 2021)*, 2021

Sound diffusing metasurfaces based on elastic plates and membranes · *European Congress and Exposition on Noise Control Engineering (EuroNoise 2021)*, 2021

Application of metamaterials to control noise scattering during space vehicle lift-off · *European Congress and Exposition on Noise Control Engineering (EuroNoise 2021)*, 2021

Launch sound level characterisation and mitigation: numerical modelling framework and metamaterial proof of concept · *16th European Conference on Spacecraft structures, Materials & Environmental Testing*, 2021

Acoustic field prediction during the launch of rockets · *Tecniacústica 2020: 50º Congreso*

Open-source GitLab MCP server that exposes 850+ GitLab actions (1,000+ depending on edition and deployment) to AI assistants through a dynamic 2-tool find/execute design, with stdio/HTTP/OAuth transports and safe & read-only modes.

**Kleidos** C++ · ESP-IDF · Embedded security
In development
Hardware password manager firmware for nine M5Stack and LilyGO ESP32 boards: credentials stay encrypted on the device (AES-256-CBC with HMAC-SHA256, PBKDF2) and are typed over Bluetooth LE as a HID keyboard after an on-device unlock, with no app and no cloud.

**phonometry** Python · DSP · Acoustics
132 ★ • 28 releases • PyPI
Standards-conformant acoustics library: sound level metrology, psychoacoustics, room, building, environmental and underwater acoustics, vibration, electroacoustics and wave simulation, every method implemented from the standard, with 1897 CI-enforced conformance checks across 111 domains, against 227 normative designations and 125 further published sources. Formerly PyOctaveBand.

**cs-routeros-bouncer** Go · Firewall · MikroTik · WAF
23 ★ • 21 releases • On the CrowdSec Hub
CrowdSec bouncer for MikroTik RouterOS: feeds the collaborative IPS/WAF CrowdSec decisions into the RouterOS firewall (address lists) through its API, blocking malicious IPs in real time.

**TFG-TFM_EPS** LaTeX · Academic template
84 ★ • 6 releases
LaTeX template for Bachelor's and Master's theses at the University of Alicante (EPS), with an automated CI build pipeline.

## // EDUCATION

**PhD program in Health and Wellness Technologies (discontinued)** 2019 - 2023
Polytechnic University of Valencia · Valencia, Spain

**Master's Degree in Acoustical Engineering**
Polytechnic University of Valencia · Gandia, Spain

**Degree in Telecommunication Engineering in Sound and Image**
University of Alicante · Alicante, Spain

**Superior Sound Technician (Higher Vocational Degree)**
IES Luis García Berlanga · Alicante, Spain

**Electronics Technician (Intermediate Vocational Degree)**
IES Salesianos (San José Artesano) · Elche, Spain

*Español de Acústica. XI Congreso Ibérico de Acústica. Faro, Portugal*, 2020

Beyond Schroeder diffusers using acoustic metasurfaces · *Tecniacústica 2020: 50º Congreso Español de Acústica. XI Congreso Ibérico de Acústica. Faro, Portugal*, 2020

Aprendizaje basado en proyectos en las materias transductores acústicos y vibroacústica · *Memorias del Programa de Redes-ICE de calidad, innovación e investigación en docencia universitaria. Convocatoria 2017-18*, 2018

Cálculo corregido, basado en la teoría moderna, de los campos acústicos (directo, temprano y tardío) · *Tecniacústica 2018: 49º Congreso Español de Acústica; XI Congreso Iberoamericano de Acústica; X Congreso Ibérico de Acústica*, 2018

Campo directo (útil)/reverberado (perjudicial) resultados experimentales frente a simulación en EASE · *Tecniacústica 2018: 49º Congreso Español de Acústica; XI Congreso Iberoamericano de Acústica; X Congreso Ibérico de Acústica*, 2018

Comportamiento vibroacústico de contenedores cilíndricos en aire · *Tecniacústica 2018: 49º Congreso Español de Acústica; XI Congreso Iberoamericano de Acústica; X Congreso Ibérico de Acústica*, 2018

Campo directo (útil)/reverberado (perjudicial) resultados experimentales frente a simulación en catt-acoustic · *Tecniacústica 2017: 48º Congreso Español de Acústica; Encuentro Ibérico de Acústica; European Symposium on Underwater Acoustics Applications; European Symposium on Sustainable Building Acoustics*, 2017

### Theses

Difusores acústicos basados en resonadores de membrana y placa · *Universitat Politècnica de València. Departamento de Física Aplicada. Escuela Politécnica Superior de Gandia*, 2019

Estudio de la relación campo directo/reverberado; útil/perjudicial · *Universidad de Alicante. Departamento de Física, Ingeniería de Sistemas y Teoría de la Señal*, 2018